Prep Test 4: Functions

75 minutes | 90 points | 11 questions

Name: _____ Date: _____

Show your work. Label axes, intercepts, and requested points. Use interval notation where requested.

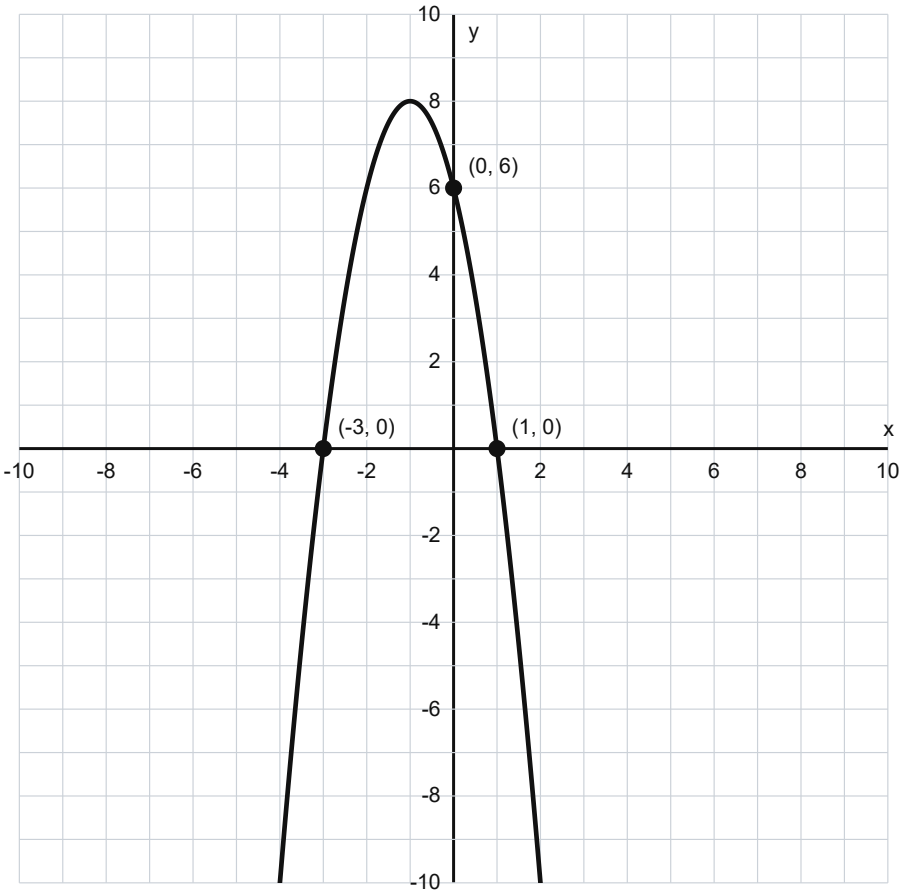
A four-function calculator and one single-sided notesheet (no example problems) are allowed. No graphing calculator.

Use the provided blank space for answers. Both axes on each grid have matching negative and positive ranges. Each square represents one unit. Curves may continue beyond the grid.

Question 1

6 points

Write an equation for the graphed function in factored and standard form. Show your work.



Given graph

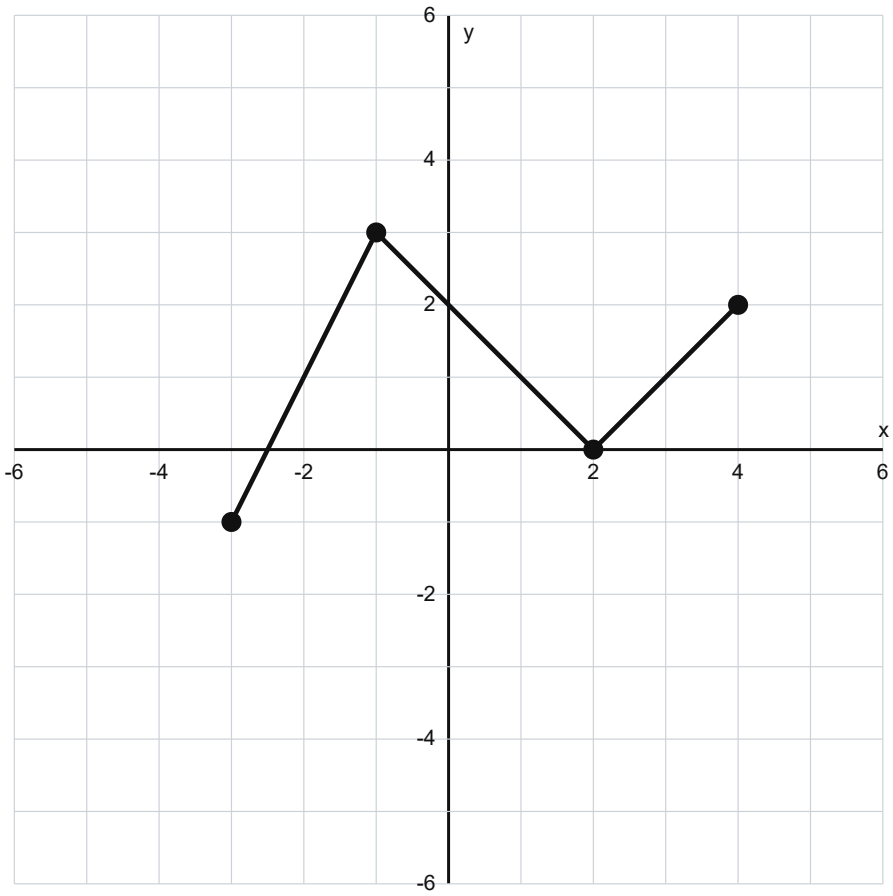
Question 2

14 points

The graph of f consists of straight segments through (-3,-1), (-1,3), (2,0), and (4,2), in that order, with included endpoints. Let g(x)=1-(3/5)x.

(a) Graph $f(g(x))$ and $g(f(x))$ on the provided coordinate plane. Use a solid line for $f(g(x))$ and a dashed line for $g(f(x))$. Label the vertices and both graphs.

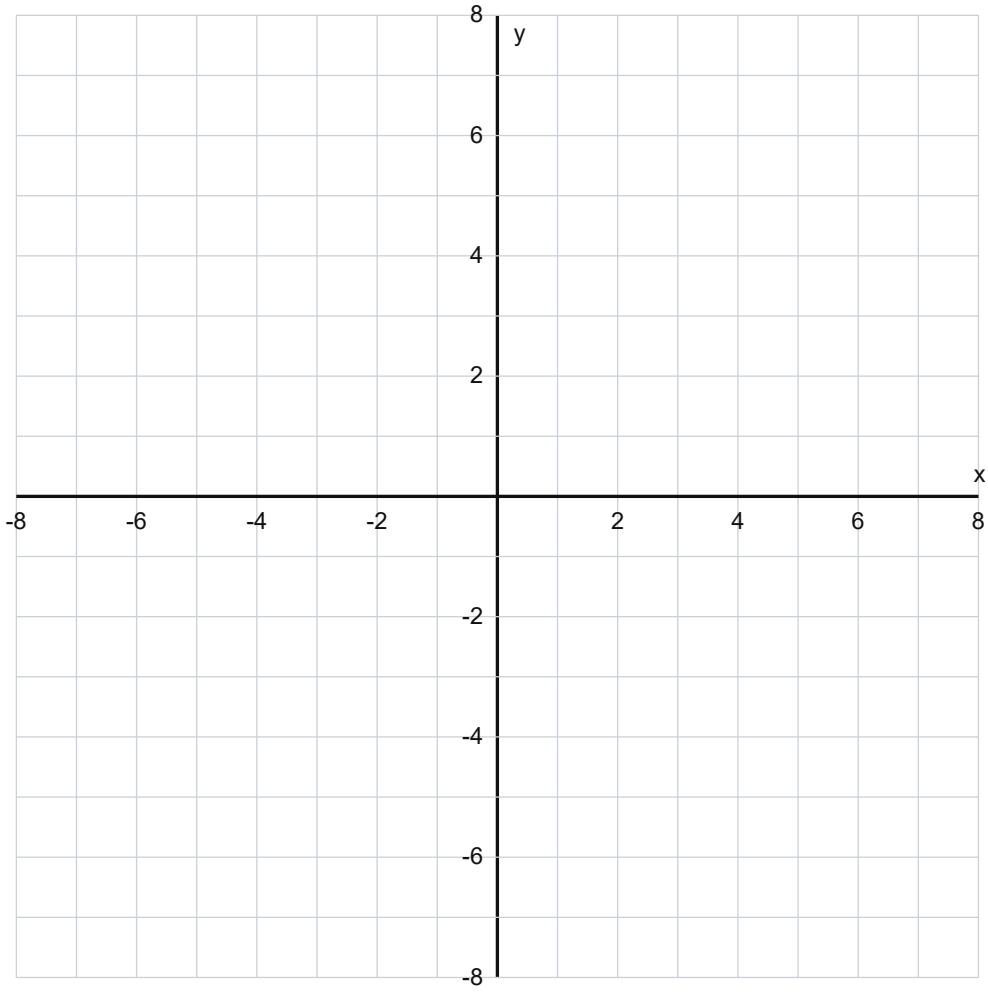
(b) Give both domains and describe the transformations of f in each composition. Show your work.



Given graph

Question 2 (continued)

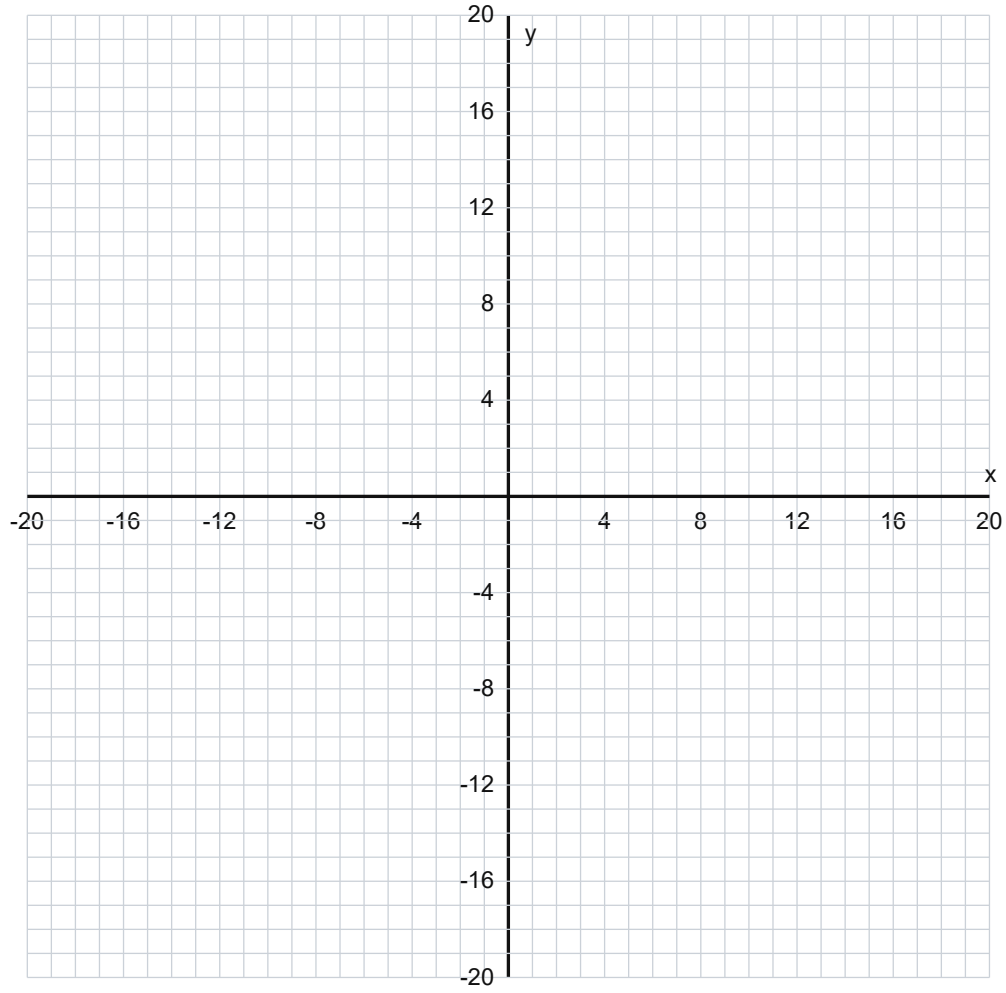
Graph both compositions here. Show coordinate work, give both domains, and describe the transformations.



Question 3

8 points

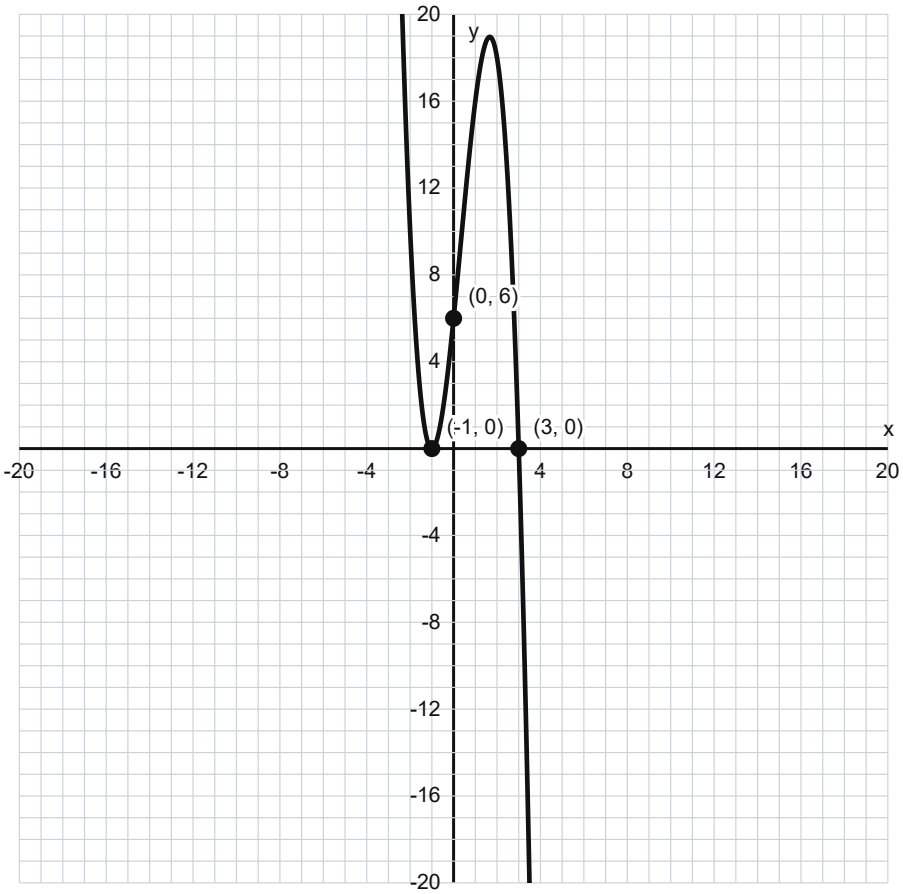
Graph $F(x) = -2x^3 + 2x^2 + 12x$. Label the axes and intercepts.



Question 4

8 points

Write an equation for the graphed polynomial. Show your work.



Given graph; curve continues beyond the grid

Question 5

10 points

Let $f(x) = \sqrt{15 - 3x}$ and $g(x) = x + 2$.

(a) Find $\frac{f(x)}{g(x)}$ and its domain.

(b) Find $g(f(x))$.

(c) Find $f(a + b)$.

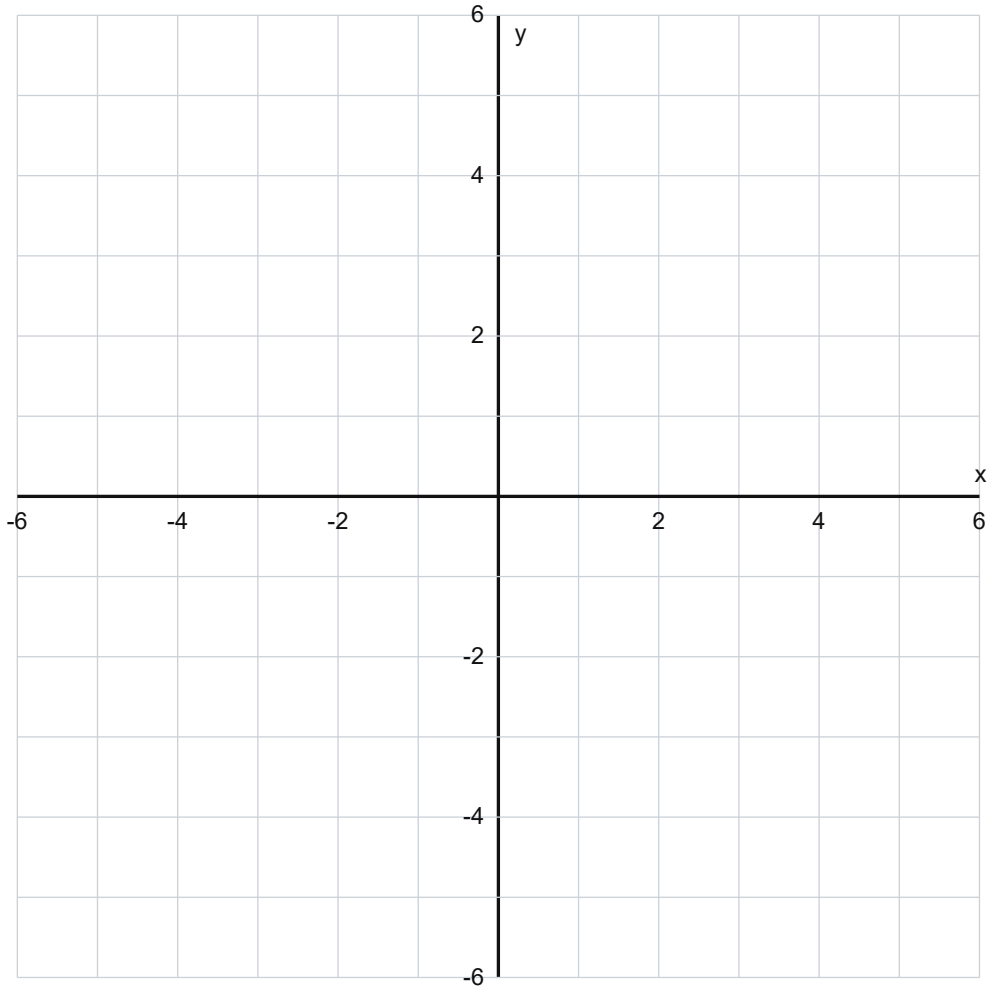
Question 6

8 points

Let $p(x) = \begin{cases} 2(x-1)^2 - 3, & x < 1 \\ -x + 2, & x \geq 1. \end{cases}$

(a) Graph p on the provided axes.

(b) Find $p(0)$, $p(1)$, and $p(3)$.



Question 7

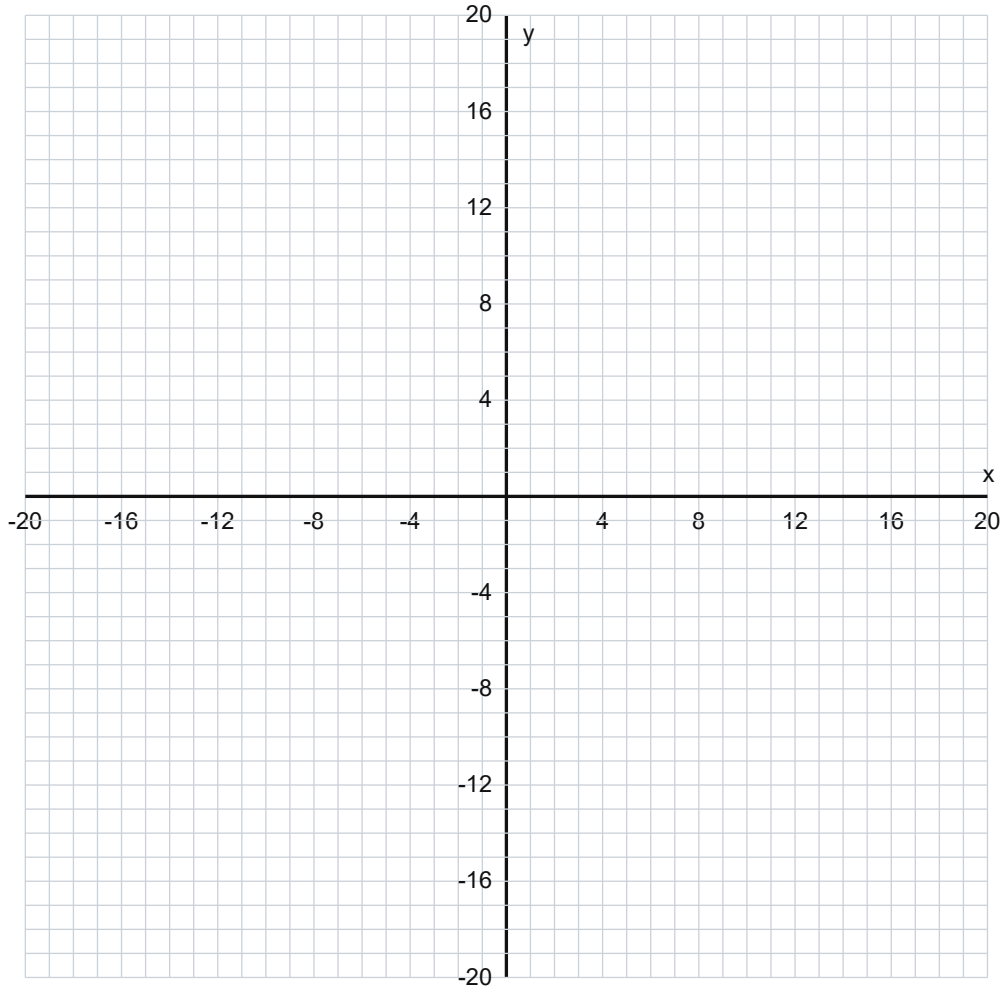
8 points

Let $R(x) = -(x+2)(x-1)^2(x-3)$.

(a) State the degree, leading coefficient, zeros, and multiplicities. Find the y-intercept.

(b) Sketch R on the provided axes. Label its intercepts.

(c) State the intervals where $R(x) > 0$ and $R(x) < 0$. Explain why its ends point the same way, unlike the odd-degree polynomial in Question 3.



Question 8

10 points

A rectangular planting bed has perimeter 24 feet. Its width is w feet, with $2 \leq w \leq 6$, and its length is $12 - w$ feet. Its area is $A(w) = w(12 - w)$ square feet.

Soil delivery costs $C(a) = 8a$ dollars when $0 \leq a \leq 30$ square feet, and $C(a) = 240 + 5(a - 30)$ dollars when $a > 30$ square feet. There are no other charges.

(a) Write the composition giving cost in terms of width. Explain the order using units.

(b) Write that cost as a piecewise function of w, including the width intervals for each price rule. State the model domain.

(c) Find the cost when w=4.

(d) With a budget of $260, find the greatest permitted width. Give an exact value and a decimal to the nearest hundredth. Show your work.

Question 9

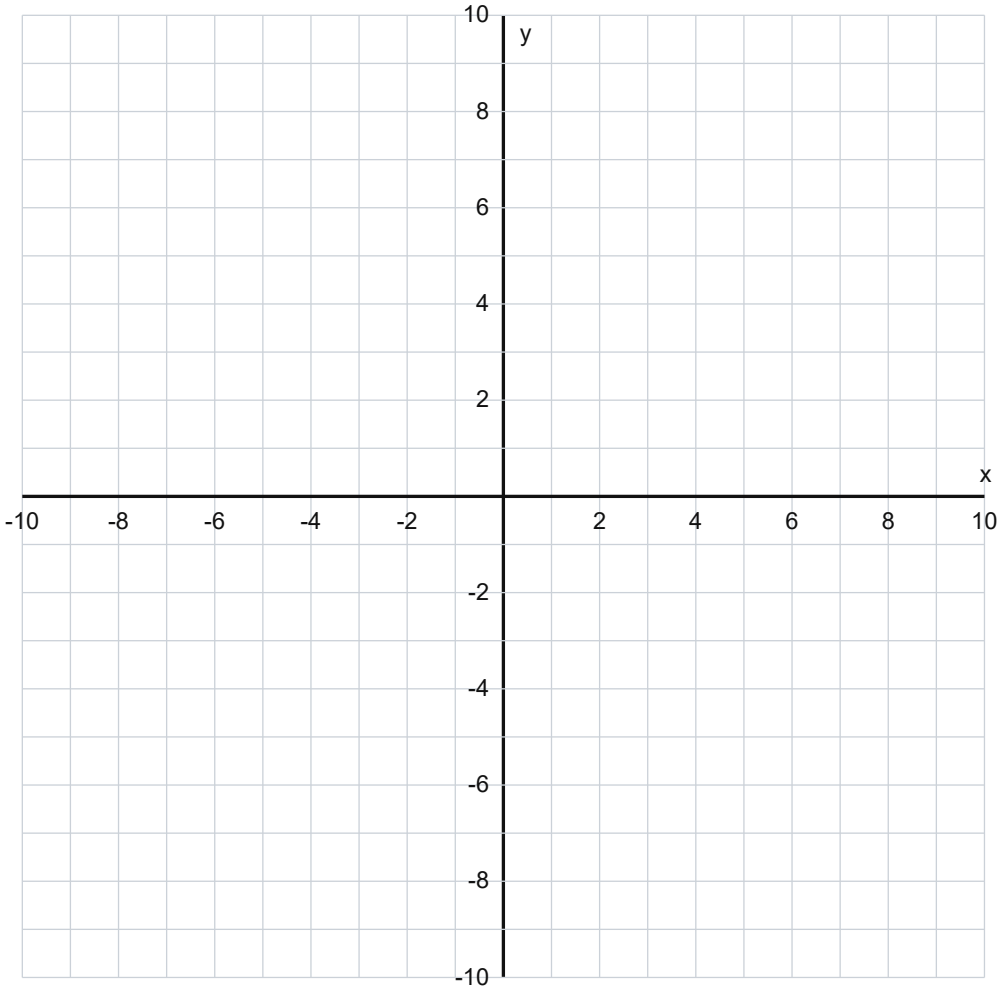
8 points

A square-root function has the form $r(x) = a\sqrt{x-h} + k$, domain $[1, \infty)$, range $(-\infty, 3]$, and passes through $(5, -1)$.

(a) Find its equation. Show your work.

(b) Graph r on the provided coordinate plane. Label its starting point and three other points.

(c) State its x-intercept exactly and its interval of decrease.



Question 10

4 points

The continuous function g is defined for all real numbers. The table gives some values and its behavior on the indicated intervals.

Which statement is guaranteed to be true?

(A) g has a relative maximum at x=-1.

(B) The range of g is [-2,6].

(C) g(x)=0 for some x in (4,8).

(D) g has a relative minimum at x=4.

Values and behavior of g

x	**-5**	**(-5,-1)**	**-1**	**(-1,4)**	**4**	**(4,8)**	**8**
g(x)	6	decreasing	-2	increasing	3	decreasing	-2

Question 11

6 points

The graph of g is shown for $-6 \leq x \leq 6$. Selected values of f are in the table; f is defined for all real numbers. Let $h(x) = 2x - 5$.

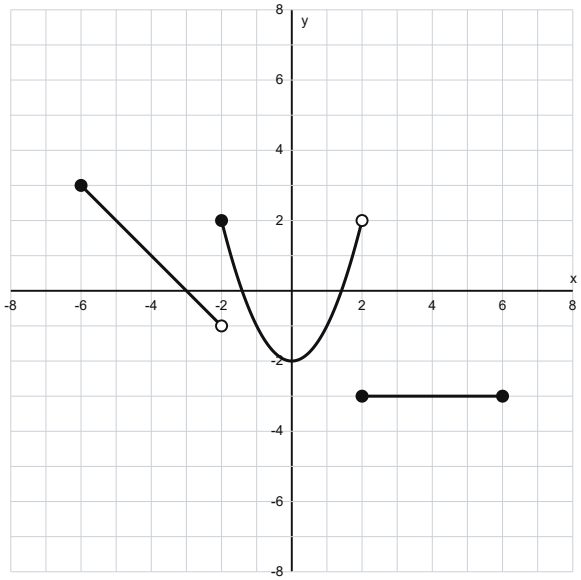
(a) Find $f(g(-5))$.

(b) Let $j(x) = g(x) - h(x)$. Find $j(2)$.

(c) What is the domain of $\frac{f(x)}{h(x)}$?

Selected values of f

x	-3	-1	0	2	4
f(x)	6	-2	3	7	-5



Given graph